

Math 110 Fall 2026 Running Review Document

Ethan Ebbighausen

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1 Disclaimer

This review document will be based on the lectures of Alan Hammond in the Fall of 2026 (our course). These lectures are themselves based on the lectures of Edward Frenkel, and Sheldon Axler's *Linear Algebra Done Right*. These are supplementary and are not a replacement for the lecture notes, or for attending lecture. They are meant as a short recap with some guiding questions based on my personal understanding of the content.

2 Sets and Fields

A set, in its essence, is just some collection of objects. That is the definition we use in discrete math, and for many purposes, though it is a naive definition (Renee Descarte and several others brought up paradoxes in this basic idea, leading to the creation of the Zermelo-Fraenkel axioms as the foundation of set theory). Most of the set theory we need is a review of Math 55, Discrete mathematics. You should be comfortable using set notation, like saying $x \in M$ for “ x is an element of M ”, or using set-builder notation like $\{(x, x^2) \mid x \in \mathbb{R}\} \subset \mathbb{R}^2$ (*what is this set?*). You should also recognize unions, intersections, complements, set differences, and cartesian products. This is also a good time to review the basics of proofs and generally interesting spaces, like the natural numbers $\mathbb{N}$, the integers $\mathbb{Z}$, and the real numbers $\mathbb{R}$.

- 1.) Prove that $(\bigcup_{a \in A} U_a)^C = \bigcap_{a \in A} (U_a^C)$. (This is DeMorgan's law)
- 2.) Prove that, for an integer x , x^2 is even if and only if x is even (recall that for if and only if, you need to make two sub-arguments where you assume different things).
- 3.) Define injective, surjective, and bijective. Construct a function which is injective but not surjective. Construct another that is surjective but not injective.
- 4.) Construct a function $f : \mathbb{N} \rightarrow \mathbb{Z}$ to show that these two sets have the same cardinality.

Fields are a special type of set. In particular, they have many additional structures to them, built to generalize the properties of the real numbers. In particular, they are a *unital ring* in which every nonzero element has a multiplicative inverse. Let's break down this definition. The “ring” part means that addition and multiplication are defined on the elements of the ring (where addition is commutative and associative, and multiplication is associative and distributive). This addition also has an additive identity 0 so $0 + r = r$ for a ring element r , as well as an additive inverse $-r$ so $r + (-r) = 0$. Next, “unital” means we have a multiplicative identity 1 so $1(r) = r$ for every element r of our field F , and the multiplicative inverse means for each r that is not zero, we have $\frac{1}{r}$ so $r \frac{1}{r} = 1$. These fields are very important in abstract algebra, so it is worth looking over the full axiomatic definition a few times to understand it. We will mostly be working over the fields $\mathbb{R}$ and $\mathbb{C}$ (the complex numbers), though we will retain some abstraction. The complex numbers are particularly important because they are what is called an *algebraically closed* field, which just means that every polynomial with complex numbers as coefficients has all of its roots in the complex numbers. This fact is called the Fundamental Theorem of Algebra (a proof of it often relies on a set of theory called *complex analysis*, which is a bit beyond this course). You should recall how to multiply, add, conjugate, and divide by complex numbers.

- Which of the following are fields? $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{R} \times \mathbb{R}$ (is multiplication defined here?), $\{0, 1\}$
- Consider $\mathbb{Z}_3$, which is the set $\{0, 1, 2\}$ where addition and multiplication are defined modulo 3 (so $1 + 2 = 0$, $2 + 2 = 1$, $1 + 1 = 2$). Is this a field?
- Let $a = 1 + 3i$ and $b = 4 + 5i$. Compute

$$\overline{a} \frac{1}{b}$$

- Show

$$\left(\frac{-1 + i\sqrt{3}}{2} \right)^3 = 1$$

- Show $0a = 0$ for all $a \in \mathbb{F}$. *Hint:* $0 = 0 + 0$
- Show that if $1 \in \mathbb{F}$ has additive inverse -1 , then $(-1)a = -a$ for all $a \in \mathbb{F}$.

The “standard” structure we will use from this point on and compare many things to is $\mathbb{F}^n$, where $\mathbb{F}$ is a field. This is the set of n -tuples or lists $\{(a_1, \dots, a_n) \mid a_i \in \mathbb{F}\}$. Addition is defined component-wise, so

$$(a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n)$$

and we define a *scalar multiplication* by elements of the field: for $\alpha \in \mathbb{F}$,

$$\alpha(a_1, \dots, a_n) = (\alpha a_1, \dots, \alpha a_n)$$

- 1.) What is $\mathbb{R}^3$?
- 2.) Show that this multiplication is distributive, i.e. $\lambda(a + b) = \lambda a + \lambda b$ for $\lambda \in \mathbb{F}$, and $a, b \in \mathbb{F}^n$.
- 3.) Show $1x = x$ for $x \in \mathbb{F}^n$.

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